THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

FORM FOUR PRE-KIPSSA EXAMINATION

031 PHYSICS

Time: 3 Hours **2025**

Instructions

1. This paper consists of sections **A, B** and **C** with total of eleven (11) questions.
2. Answer all questions from section **A** and **B** and any two (2) questions from section **C.**
3. Where necessary, the following constants may be used
   - Acceleration due to gravity = $10ms^{-2}$.
   - Density of water = $1.0gcm^{-3}$ or $1000kgm^{-3}$

SECTION A (16 MARKS)

Answer all questions

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter beside the item number in the booklet provided.

   (i) Which of the following best describes the primary goal of Physics as a science?

   A. To create complex mathematical equations
   B. To describe and understand natural phenomena
   C. To conduct chemical experiments
   D. To manufacture technological devices
   E. To deal with how fundamental laws work

   (ii) Which of the following factors does NOT affect the pressure exerted by a gas in a closed container?

   A. Temperature of the gas
   B. Volume of the container
   C. Number of gas molecules
   D. Color of the gas
   E. None of the above

   (iii) According to Archimedes' principle, what determines the buoyant force acting on a submerged object in water?

   A. The weight of the object
   B. The volume of water displaced by the object
   C. The size of the object
   D. The density of the object
   E. Density of fluid

   (iv) Colours are produced when light passes through a glass prism because:

   A. Light waves interfere
   B. Glass prism colours the light
   C. In glass, different colours travel at different speed
   D. Different colours are filtered
   E. Diffraction of light occurs

v. Radio set rated 400w is switched on for 10 hours every day. How much energy does it consume in 15 days?

A. 21600 KJ
B. 2160,000 KJ
C. 216 KJ
D. 216000 KJ
E. D216,000,000 KJ

(vi) What does relative humidity measure?

A. The amount of moisture in the air compared to the maximum amount it can hold at a certain temperature
B. The total amount of water vapor present in the atmosphere
C. The temperature at which air becomes saturated with moisture
D. The density of water vapor in the air
E. The temperature at which the air becomes saturated with water vapor

(vii) Which layer of the earth is primarily composed of molten rock?

A. Crust
B. Mantle
C. Inner core
D. Outer core
E. No correct answer

(viii) What is the main reason for the changing phases of the Moon?

A. Earth's rotation
B. The distance between Earth and the Moon
C. The reflection of sunlight on the Moon
D. The changing distance of the Moon from the Earth
E. Force gravity between the moon and the earth

(ix) What phenomenon causes a prism to separate white light into its component colors?

A. Reflection
B. Refraction
C. Diffraction
D. Dispersion
E. Relegation

(x) Which method is preferred to use if a student wishes to charge an uncharged body by using a positively charge body in order to make it acquire a positive charge?

A. Friction

B. Contact

C. Induction

E. convection

D. Heating

2. Match the items in list **A** with responses in list **B** by writing the letter of the correct response beside.

LIST A	**LIST B**
i. Specific heat capacity ii. Heat capacity iii. Latent heat of fusion iv. Latent heat v. Regelation vi. Evaporation	A. The energy possessed by object due to its temperature B. The phenomena of melting and re-freezing under the influence of pressure C. The process through which the liquid changes to vapor (gas) at a temperature even below its boiling point. D. It is the vaporization process where liquid changes into gas system when heated E. Refers to the energy absorbed or released by a substance during a phase change, such as melting, freezing, boiling, or condensing, without any change in temperature F. The amount of heat energy required to raise the temperature of one-unit mass of a substance by one degree Celsius (or one Kelvin). G. The amount of energy required to change a substance from solid to liquid (or vice versa) at a constant temperature. H. The total amount of heat energy required to change the temperature of an entire object or system by one degree Celsius (or Kelvin).

SECTION B (54 MARKS)

Answer all questions in this section

3. (a) The dispersed beam of light (seven colours) can recombine to form single white light. Using two glass prisms, sketch the diagram to show how this is formed.

   (b) A rod of length 10 cm long lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. What is the length of the image?

4. (a)(i) A hole at the bottom of a ship is more dangerous than one near the surface. Explain why?

   (ii) Briefly explain, why heavy trucks are made up with many tyres?

   (b) A uniform rod with a mass of 120g and length 130 cm is suspended by a wire from a point 80 cm from the rod's left end.

   (i) What mass must be hung from the right end of the rod for it to be in equilibrium?

   (ii) What will be the tension in the wire?

5. (a) Use Newton's 2nd law of motion to deduce that impulse is equal to the change in the linear momentum of a body and explain why it is necessary to fold our legs when we jump from a height on landing.

   (b) A piston of hydraulic press has their areas given as $0.0003m^2$ and 2 cm^2 respectively. I20N is required to push down the small piston. Find the force required to push the large piston

6. (a) A beaker containing water ice is heated from -5^0C to 0^0C and then from 0^0C to 15^0C. With the aid of a diagram, explain the variation of density with temperature.

   (b) A heater of 500w is used to raise the temperature of 50 kg of material of specific heat capacity of 960 J/kg K, from 18^0C to 38^0C. Assume that all the heat from the heater is given to the material, calculate:

   (i) Heat capacity of the material

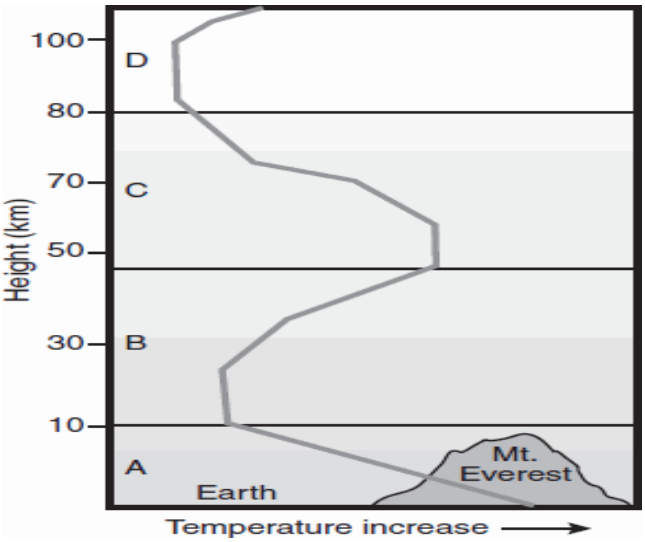
   (ii) The time taken in seconds

7. (a) A duck remains floating at the same place as waves pass by the water in a lake. Explain

   (b) A certain radioactive material has a half-life of 2 minutes. If its initial count rate is 256 per minute, determine the fractional number of atoms left un decayed after 6 minutes.

8. (a) What is the difference between gravity and gravitational force and briefly explain how they are related.

(b) Study the vertical structure of the atmosphere in the figure below and then answer the asked questions.



(i) The temperature in the layer A decreases with an altitude at average rate of 6.5^0C/km. What is the reason for this decrease**?**

(ii) Why layer D is called communication layer?

SECTION C (30 Marks)

Answer only TWO questions from this section

9. (a) (i) Faraday's law and Lenz's law are precise laws dealing with induced a.m. How do they differ when there is magnetic flux linking the coil concerned?

(ii) The results of Faraday's experiments were that magnitude or the induced e.m.f were related to which factors? Explain in short four factors.

(b) An ideal transformer should have an efficiency of 100%. Yet in any practical transformer there are power losses which do not allow 100% efficiency. Explain five of those factors.

(c) A step-up transformer has I 0 000 turns in the secondary coil and 100 turns in the primary coil. A current of 5.0A flows in the primary circuit when connected to a 12.0V supply.

(i) Calculate the voltage across the secondary coil.

(ii) If the transformer has an efficiency of 90%, what is the current in the secondary coil?

10. (a) Two wires A and B are made of the same material. A has half the length and twice the diameter of B. What is the ratio of the resistance of B to that of A?

    (b) State the defects of a simple cell and explain how these defects may be minimized?

    (c) Robert complains that one of her wall sockets is not functioning properly. He checked live and neutral terminals using a domestic electric tester. Both indicate the presence of electricity, but the appliance does not work. Give a possible explanation.

11. (a)A helicopter is hovering at an altitude of 200m above the surface of a lake. A speaker on the helicopter is sending out sound waves, which are reflected from both the surface of water and bottom of the lake. If the difference in arrival times of the two echoes is measured to be 0.24 seconds, what is the depth of lake? (The atmospheric temperature is 20^0C).

    (b) (i) Draw a labeled diagram of a simple d.c motor

    (ii)Explain why the coil will rotate when the current flows in it and why it will turn in a specific direction**.**

    (iii) Why does such motor occasionally not start to rotate when the current is switched on but will give continuous rotation after a slight push?